

4.3 TRUSS DISPLACEMENTS

4.3.1 The Deformed Shape

When a truss is loaded, each member will be subjected to a tensile or compressive force. As a result of these forces, each member will suffer elastic strain leading to a change in the original length of the member. The members in tension will become longer while the members in compression become shorter. However, compatibility at the joints requires that the members remain joined together. In order to satisfy equilibrium, compatibility and constitutivity, the joints which are not restrained will be displaced from their original positions, so that the truss assumes a slightly different geometry. The geometry adopted by the truss after loading is referred to as the **deformed shape**.

The member elongations or shortenings are usually very small in comparison to their original lengths. However, the cumulative effect of the member straining can lead to a significant change in the original geometry of the truss.

Structural analysis aims at predicting and evaluating the joint displacements since they affect certain design criteria. In this section, the **Virtual Unit Load** method shall be applied to determine joint displacements in plane and space trusses.

4.3.2 The Particular case

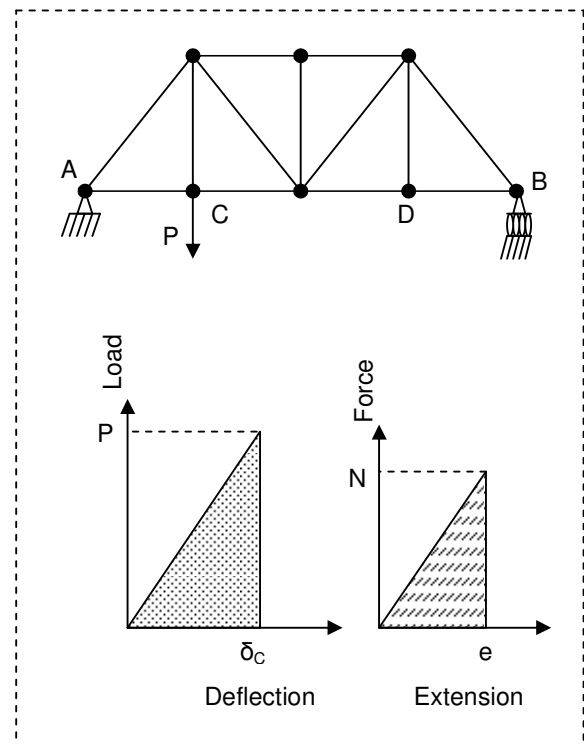
Suppose it is required to determine the downward deflection of the joint C in the truss shown below.

Assuming the load P is gradually applied, we can plot the load-deflection curve as shown.

The external work done is the given by:

$$\text{Ext. W.D} = \frac{1}{2}P\delta_C$$

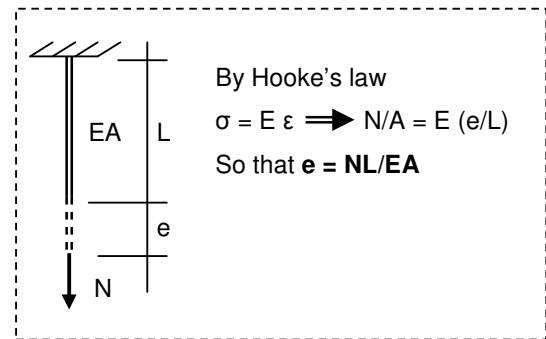
Let the force induced in each member be N . Each member suffers an axial extension given by $e = NL/EA$ as shown below



We can plot of the force-extension curve for any member as shown.

The internal work done on any member is given by: $\text{Int. W.D} = \frac{1}{2}Ne = \frac{1}{2}N \text{ NL}/EA$
 $= N^2L/2EA$

The total internal work done is obtained by summation thus: $\text{Int. W.D} = \sum N^2L/2EA$



The principle of conservation of energy dictates that $\text{Ext. W.D} = \text{Int. W.D}$

$$\frac{1}{2} P \delta_C = \sum N^2L/2EA \quad \text{-----} \quad \text{Eqn 3}$$

Eqn 3 above may be used to evaluate δ_C .

Note: In Eqn 3, the required deflection δ_C is the deflection of the point C, which is the point of application of the load P, which is the only applied load. Hence Eqn 1 is called **the particular solution**.

4.3.3 The General Case

Suppose it was required to determine the downward deflection of point D.

The solution may be achieved in four steps:

Step 1: Imagine the applied load was absent.

Apply a virtual unit load at the point D, in the direction and sense of interest.

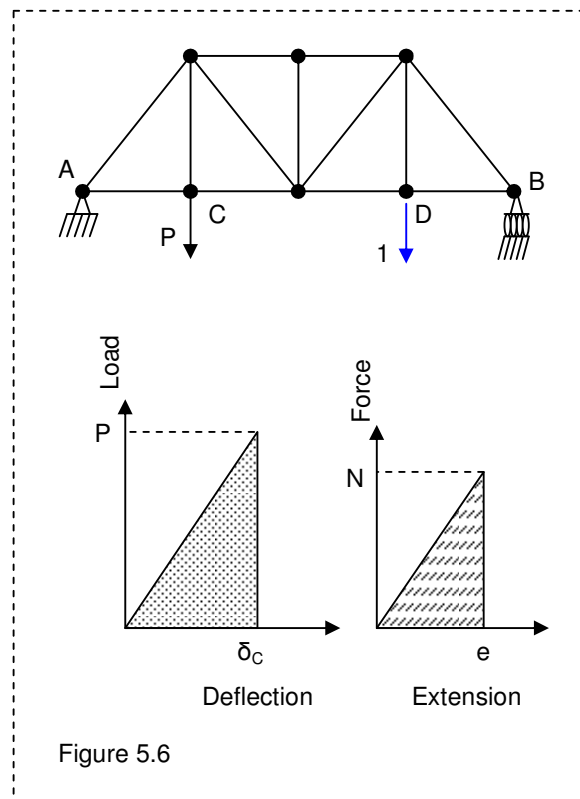
The virtual unit load would induce virtual displacements which are unreal.

Step 2: With the virtual unit load in place, introduce the applied loading P, gradually from magnitude zero to its full magnitude.

As P rides through δ_C , the virtual unit load will ride through δ_D .

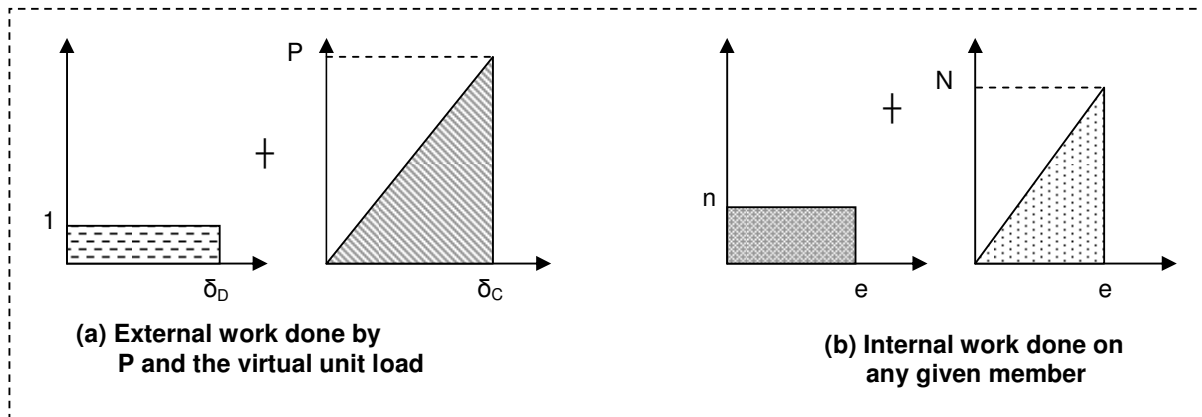
Let **n** be the axial force induced in any member due to the virtual unit load,

Let **N** be the axial force induced in any member due to the applied load.



Let $\delta_{C\downarrow}$ be the vertical downward displacement of the joint C due to the applied load P and $\delta_{D\downarrow}$ be the vertical downward displacement of the joint D due to the applied load P.

Step 3: The external work done by both P and the unit virtual load, and the internal work done on any given member are shown below.



Step 4: The principle of conservation of energy requires that:

$$\text{External Work Done} = \text{Internal Work Done} \Rightarrow 1 \cdot \delta_D + \frac{1}{2} P \delta_C = \sum n e + \sum \frac{1}{2} N e$$

But, the particular solution requires that: $\frac{1}{2} P \delta_C = \sum \frac{1}{2} N e$

Thus we deduce that: $\delta_D = \sum n e = \sum n (NL/EA)$

$$\delta_D = \sum n NL/EA \quad \text{-----} \quad \text{Eqn 4}$$

Eqn 4 represents the general solution for axial displacements, and may be expressed in general format thus:

$$\delta_i = \sum n_i (N/EA) L \quad \text{-----} \quad \text{Eqn 4*}$$

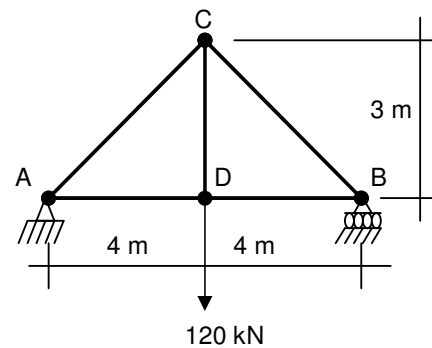
4.4.4 Solution Procedure

1. Analyse the truss under the given loading to obtain **[N]** member forces.
2. For any required i^{th} displacement: -
 - ✓ Assume the applied loading was absent.
 - ✓ Apply a virtual unit load at the joint of interest, in the direction and sense of interest
 - ✓ Analyse the truss to obtain the **[n_i]** member forces
3. Next required displacement.
4. Make a suitable table to facilitate your calculations.
5. For each required displacement, evaluate $\delta_i = \sum n_i NL/EA$
6. Next required displacement
7. Stop: END.

Example 1

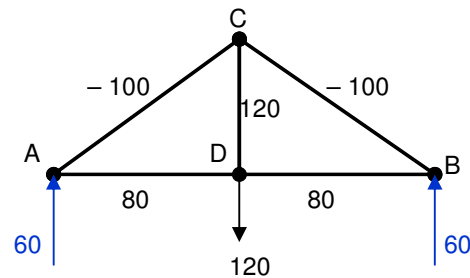
Determine $\delta_{C\downarrow}$, $\delta_{C\rightarrow}$ and $\delta_{B\rightarrow}$

Take $EA = 210 \times 10^3 \text{ kN}$ for all members.



Solution

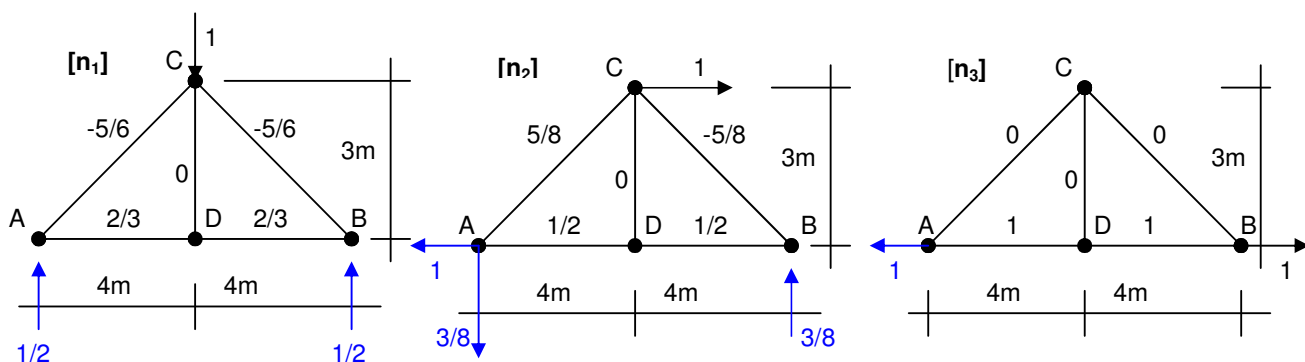
Step 1: Analyse the given truss and obtain the **[N]** member forces.



Step 2: Assume the applied load was absent. Let $\delta_1 = \delta_{C\downarrow}$. Apply a virtual unit load acting vertically downwards at C, then analyse the truss to obtain **[n₁]** member forces.

Step 3: Assume the applied load was absent. Let $\delta_2 = \delta_{C\rightarrow}$. Apply a virtual unit load at C acting horizontally to the right, then analyse the truss to obtain **[n₂]** member forces.

Step 4: Assume the applied load was absent. Let $\delta_3 = \delta_{B\rightarrow}$. Apply a virtual unit load at B acting horizontally to the right, then analyse the truss to obtain **[n₃]** member forces.



Step 5: Make a table as shown below. (Note that EA is constant and can therefore be factored out and applied in the calculation of the displacements after the summation of the other quantities has been achieved).

Tabulation

Member	L (m)	N (kN)	n_1	n_2	n_3	n_1NL	n_2NL	n_3NL
AC	5	-100	-5/6	5/8	0	1250/3	-625/2	0
AD	4	80	2/3	1/2	1	640/3	160	320
BC	5	-100	-5/6	-5/8	0	1250/3	625/2	0
BD	4	80	2/3	1/2	1	640/3	160	320
CD	3	120	0	0	0	0	0	0
Σ						1260	320	640

Step 6: Calculations

$$\bar{\delta}_1 = \bar{\delta}_{C\downarrow} = \sum n_1NL/EA = 1260/210 \times 10^3 = 6.0 \times 10^{-3} \text{ m} = 6.0 \text{ mm}$$

$$\bar{\delta}_2 = \bar{\delta}_{C\rightarrow} = \sum n_2NL/EA = 320/210 \times 10^3 = 1.5 \times 10^{-3} \text{ m} = 1.5 \text{ mm}$$

$$\bar{\delta}_3 = \bar{\delta}_{B\rightarrow} = \sum n_3NL/EA = 640/210 \times 10^3 = 3.0 \times 10^{-3} \text{ m} = 3.0 \text{ mm}$$

Note that the overall displacement of the point C may be obtained vectorially as shown in the diagram, thus:

$$\bar{\delta}_C = \sqrt{\{6.0^2 + 1.5^2\}} = 6.2 \text{ mm, inclined at an angle given by}$$

$$\theta = \tan^{-1} (6.0/1.5) = 76^\circ. \text{ However, such overall displacement}$$

is not of much interest in engineering, and it will therefore be sufficient to deal with the vertical and horizontal components of the joint displacement.

